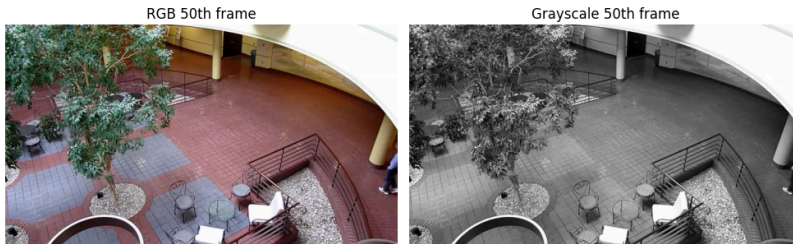


Problem 1:

When viewing the 50th frame, only one person is visible in the bottom-right at frame 50.



After taking the SoD on the sequence of frames, we get an image which is almost entirely black except a bright blob in bottom-right corresponding to the moving person.

Sum-of-Diff for frames 49-51



After detection, we get a bounding box for that blob.



Then, matches were found between the deep features extracted from these bounding boxes across consecutive frames to perform data association tracking.

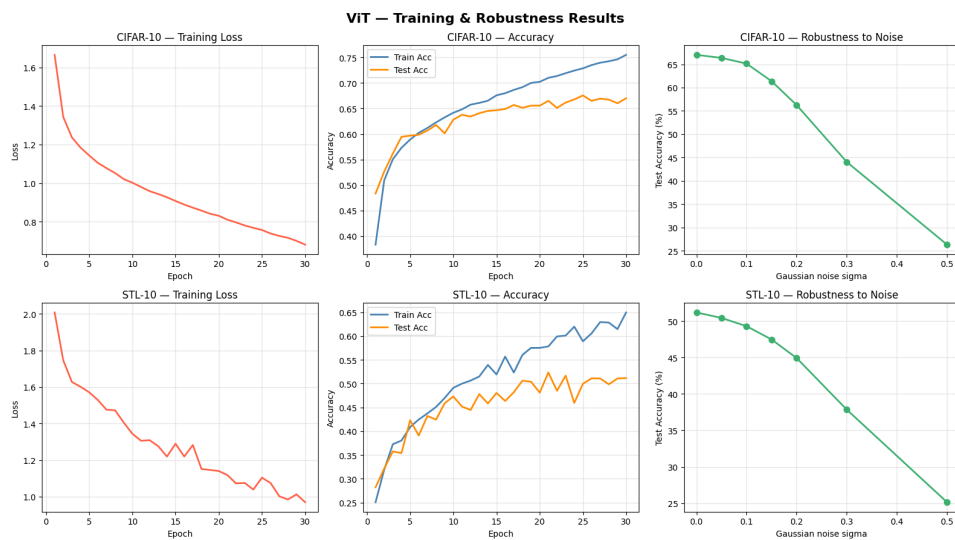


Conclusion: A bounding box was detected around only the moving person (recall is 1 and precision is 1 as well) although not around the whole person. But, we are able to consistently follow the moving person in the video in the notebook.

Problem 2:

Choosing CIFAR-10 and STL-10 because performance of multiclass classification among 10 classes in both could be used to compare domain sensitivity of the same model architecture.

Results:



CIFAR-10:

Over 30 epochs, test accuracy had an upward trend and best test accuracy was 67.6% at epoch 25. Some overfitting is happening as train accuracy has reached 75.5%. It is also mildly overfitted (a gap of ~8% in train and test acc) and pretty gaussian noise-sensitive as performance degrades a lot per sigma after a certain sigma (0.10). (noted in notebook)

STL-10:

Much noisier updates to loss and accuracies in the STL-10 data as the STL-10 train set (5000) is much smaller than CIFAR-10 (50000). Best test accuracy of 52.3% at epoch 21. For STL-10, it's strongly overfitted (gap of ~14% in train and test acc) with a lot of noisy updates while training. This is also seen as the test accuracy started dropping/stayed constant after epoch 21.

Robustness Under Gaussian Noise:

The CIFAR-10 performance started dropping steeply after $\sigma=0.10$ (67% $\rightarrow$ 44% at $\sigma=0.3 \rightarrow$ 26% at $\sigma=0.5$). STL-10 degrades more steadily in those steps (51% $\rightarrow$ 38% at $\sigma=0.3 \rightarrow$ 25% at $\sigma=0.5$) but drops steeply after $\sigma = 0.3$.

Overall, a lower accuracy was achieved on STL-10 compared to CIFAR-10. It is observed that accuracy in STL-10 doesn't drop as much with increasing noise as CIFAR's does. Thus the CIFAR-10 dataset seems more sensitive to Gaussian noise. This could be because of a resizing (96x96 $\rightarrow$ 32x32) we had to apply to STL-10 to adapt it for the same model architecture as CIFAR-10, which while removing high frequency details which could've made it more robust to noise interference.